

RAMDANI KHALED

AI Systems Builder | Analytics engineer
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PROFESSIONAL SUMMARY

Final-year Computer Science student at Université Badji Mokhtar – Annaba, specializing in Analytics Engineering and AI. Currently designing a multi-agent ML employment recommendation system. Machine learning builder driven by one goal: build AI systems with real-world impact.

EDUCATION

Bachelor of Science in Computer Science *Expected: 2026*

Universite Badji Mokhtar, Annaba, Algeria | Member, AI & Robotics Association of Annaba Wilaya

EXPERIENCE

AI Systems Intern *Jan 2026 – Present*

- Architecting an employment intelligence platform using a multi-agent ML architecture on large-scale NoSQL Big Data
- Designed a four-stream NoSQL pipeline spanning Profiles, Offers, Placements, and References using MongoDB and Elasticsearch
- Built a KNN-based recommendation engine for skill gap analysis and job-candidate matching
- Implemented dynamic weight optimization using supervised learning trained on historical placement data

PROJECTS

ARIA – Adaptive Reasoning & Intelligence Architecture | In Development *Python · LangGraph · Ollama · ChromaDB · Streamlit*

- Multi-brain AI agent (Planner, Advisor, Meta-Evaluator) with human approval checkpoint before execution
- Three-layer memory: Working Memory, Episodic Memory (ChromaDB), Strategic Memory (NetworkX knowledge graph)
- Privacy-first local inference via Ollama with optional cloud fallback; SHA-256 cryptographic audit trails per step

AI Engine – Employability Scoring System | Active · Internship *MongoDB · Elasticsearch · scikit-learn · Python*

- Multi-agent ML system for I employment analytics processing large-scale workforce records
- Per-strategy Logistic Regression models with feature importances fed back as dynamic weight overrides

Wyag Lite – Git Reimplementation in Python | Complete *Python · Systems Programming · Git Internals*

- Rebuilt Git internals from scratch in pure Python with zero external dependencies; implements commits, trees, refs, tags, and log

TECHNICAL SKILLS

AI & Agents: Multi-agent architecture, LLM orchestration, LangGraph, RAG, vector stores, local inference (Ollama), agentic pipelines

Machine Learning: scikit-learn, NumPy, Pandas, OpenCV, MediaPipe, supervised learning, KNN,

Data Engineering: MongoDB, NoSQL design, SQL

Development: Python (primary), JavaScript, C, Java (basic), Streamlit, REST APIs, Git

LANGUAGES & AFFILIATIONS

Arabic (Native) | English B1-B2 (3 years university instruction) | Member, AI & Robotics Association of Annaba Wilaya (government-recognized)